University of Perpetual Help-Dalta

Molino Campus

Student Activity: Parallel Computer Architecture

Student Information

Name: Alexander Jon S. Solis

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Learning Objectives

- Define parallel computer architecture
- Identify types of parallel architectures
- Explain real-world examples of parallel computing
- Understand the advantages of parallel processing

Activity 1: Think & Answer

1. What is parallel computer architecture?

Parallel computer architecture is a computing system design where multiple processing units are working simultaneously to solve a problem or execute tasks instead of using a single processor that handles one instruction at a time.

2. How is parallel processing different from serial processing?

Serial processing is a type of processing wherein it executes an instruction one after another in sequence, completing a task before moving on to the next while the Parallel processing, executes instructions or tasks simultaneously using multiple processors.

3. Give one real-life example of parallel processing.

A real-life example of parallel processing is video rendering because editing and applying special effects in a video uses multiple processing units in order to process the video frames simultaneously.

Activity 2: Match the Column

1. SIMD ☐
2. MIMD ☐
3. GPU ☐
4. Supercomputer ☐

a. Multiple instructions, multiple data

b. One instruction applied to many data

c. Weather forecasting

d. Graphics and AI processing

Activity 3: Identify the Type

1.A graphics card processing millions of pixels at the sametime

Type: SIMD

2.A multi-core CPU running multiple applications

Type: MIMD

3. Networked computers solving a problem together

Type: DISTRIBUTED COMPUTING

Activity 4: Scenario-Based Question

A weather forecasting system uses thousands of processors at the sametime.

1. Is this parallel computing? ☐ Yes ☐ No

2. What type of parallel architecture is used?

The type of parallel architecture that is used in a weather forecasting system is a MIMD or the Multiple Instruction, Multiple Data architecture which is typically implemented using a supercomputer or a cluster system.

Activity 5: Reflection

Why is parallel computer architecture important today?

Parallel computer architecture is importand and crucial in today's world because most modern problems and systems require massive computational power that single processors cannot provide like artificial intelligence, simulations, big data and real-time applications. As processor speed improvements have slowed down, parallel processing has become the primary way to achieve faster computing, making it essential for technological advancement and solving complex real-world challenges.